

BTMACS201

B.Tech./B.Tech.+M.B.A./ B.Tech +M.Tech.(CSE, CSE-CF, CSE-BDA, IT, CSE-MA, BDCE-Impetus, CSE-MA, CSE-ES, CSE-BDAI, CSE-AII, CSE-CMCI, CSE-FSDI, CSE-DSI, CSE-AIML)

II Semester Examination, December 2023

Mathematics-II

Choice Based Credit System (CBCS)

Time: 3 Hrs.

Maximum Marks: 60

Minimum Pass Marks:24

- Note: (1) All questions carry equal marks, out of which part 'A' and 'B' carry 3 marks and part 'C' carries 6 marks.
 (2) From each question, part 'A' and 'B' are compulsory and part 'C' has internal choice.
 (3) Draw neat diagram, wherever necessary.
 (4) Assume suitable data wherever necessary.

Q.1(A) Find the Eigen values of the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$. 03

(B) Determine the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix}$. 03

(C) Show that the following system is consistent and solve it: 06

$$x + y + z = 3$$

$$x + 2y + 3z = 4$$

$$x + 4y + 9z = 6.$$

OR

Diagonalize the matrix $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$.

Q.2(A) Solve: $3e^x \tan y \, dx + (1 - e^x) \sec^2 y \, dy = 0$. 03

(B) Solve: $\sec x \frac{dy}{dx} = y + \sin x$. 03

(C) Solve: $p^2 + 2py \cot x = y^2$. 06

OR

$$\text{Solve: } \frac{d^2y}{dx^2} - \frac{dy}{dx} - 6y = e^x + \cos x.$$

Q.3(A) Prove that $e^x \left(\frac{\Delta^2}{E} \right) e^x \cdot \frac{Ee^x}{\Delta^2 e^x}$. 03

(B) Express $y = 2x^3 - 3x^2 + 3x - 10$ in a factorial notation and hence show that $\Delta^3 y = 12$. 03

Contd.....

- (C) From the given table find for what value of x when $y = 13.6$:

x :	30	35	40	45	50
y :	15.9	14.9	14.1	13.3	12.5

06

OR

Find approximate value of $\int_0^1 \frac{1}{1+x^2} dx$ by Simpson's one third and hence find the error from exact value of integral.

- Q.4.(A)** Find $y(2.2)$ using Euler's method from the equation: $\frac{dy}{dx} = -xy^2$ with $y(2) = 1$, take $h = 0.04$. 03
- (B) Using Euler's modified method find y at $x = 0.5$ such that $\frac{dy}{dx} = x + y^2$, given $y(0) = 1$. 03
- (C) Apply Runge-Kutta fourth order method to find y when $x = 0.4$, where $\frac{dy}{dx} = \frac{y-x}{y+x}$, given $y(0) = 1$. 06

OR

Using Milne's method to find $y(0.3)$ from $\frac{dy}{dx} = x^2 + y^2$; $y(0) = 1$. Find the initial values $y(-0.1)$, $y(0.1)$ and $y(0.2)$ from the Taylor series method.

- Q.5.(A)** Define continuous and discrete random variables. 03
- (B) An experiment consists of observing the sum of the dice when two fair dice are thrown find the probability that the sum is 7. 03
- (C) The mean height of 500 students is 151cm and the standard deviation is 15cm assuming that the heights are normally distributed, find how many students have heights between 120 and 155cm ? (from the normal table $P(0 < z < 2.07) = 0.4808$ and $P(0 < z < 0.27) = 0.1064$). 06

OR

Consider a system that can be in one of two possible states $S = \{0,1\}$. Suppose that the

transition matrix is given by $P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{3} & \frac{2}{3} \end{bmatrix}$ and the system is in state 0 at time $n = 0$, i.e., $X_0 = 0$.

Draw the state transition diagram and find the stationary distribution.

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